Oxford College of Emory University

Chapter 13: Fiscal Policy

Government Purchases, Taxes, and Transfers

ECON_OX 112 · Principles of Macroeconomics

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The Federal Budget: FY2025

Federal government only · October 2024–September 2025 · Billions of dollars

Revenues	$ billions
Individual income taxes	2,656
Payroll taxes	1,748
Corporate income taxes	452
Customs duties (tariffs)	195
Excise taxes	106
Estate and gift taxes	29
Federal Reserve remittances	5
Miscellaneous fees and fines	42
Total revenues	**5,235**

Outlays	$ billions
Social Security	1,575
Medicare (net)	988
Medicaid	668
Income security programs	397
Other mandatory (net)	539
Defense discretionary	893
Nondefense discretionary	980
Net interest	970
Total outlays	**7,010**

Budget balance: $5,235 − $7,010 = −$1,775 billion (deficit)

A financing shortfall—not a measure of the effect on GDP.

Source: CBO, FY2025 Federal Budget; Revenues; 2026 Outlook. Components may not sum because of rounding. Medicare and other mandatory outlays are net of offsets.

What Does Federal Spending Pay For?

> **Public services**
>
> National defense, federal courts, and law enforcement.

Source: CBO, Federal Budget and spending breakdowns, FY2025

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Retirement and disability benefits, food assistance, and refundable tax credits.

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Retirement and disability benefits, food assistance, and refundable tax credits.

Interest on federal debt

Payments to lenders; these do not buy current goods or services.

Source: CBO, Federal Budget and spending breakdowns, FY2025

Roadmap

Policy goals and instruments

Purchases, household taxes, and transfers

Demand and policy limits

Expansion, contraction, criticisms, and delays

The multiplier

Initial spending and successive consumption rounds

Stabilizers and budget balances

Automatic responses, deficits, and sustainability

Fiscal Policy and Stabilization

Definition

Government decisions about purchases, taxes, and transfers.

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Stabilization goal

Support output and employment in a recession; restrain excess demand in a boom.

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Support output and employment in a recession; restrain excess demand in a boom.

Policy direction

Expansionary policy raises AD; contractionary policy reduces AD.

Public services and redistribution are also fiscal goals.

Fiscal Instruments

Instrument	Example	Immediate channel
Purchases (G)	Road maintenance	A purchase of current output
Transfers (TR)	Unemployment benefits	Household disposable income
Household taxes (T)	Personal income taxes	Household disposable income

Transfers are not G. Public infrastructure investment is G; I is private investment.

GDP accounting: BEA

Fiscal Policy and Aggregate Expenditure

$$Y = \mathbf{C} + \mathbf{I} + \mathbf{G} + \mathbf{NX}$$

C: Consumption

Household taxes and transfers change disposable income.

Tax cuts or higher transfers can raise C.

I: Private investment

G: Government purchases

NX: Net exports

Fiscal Policy and Aggregate Expenditure

$$Y = C + I + G + NX$$

C: Consumption	Household taxes and transfers change disposable income. Tax cuts or higher transfers can raise C.
I: Private investment	Business tax incentives and expected sales affect investment. Borrowing can also raise financing costs and weaken I.
G: Government purchases	
NX: Net exports	

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G: Government purchases	Government buys current goods and services. The direct expenditure channel.
NX: Net exports	

Fiscal Policy and Aggregate Expenditure

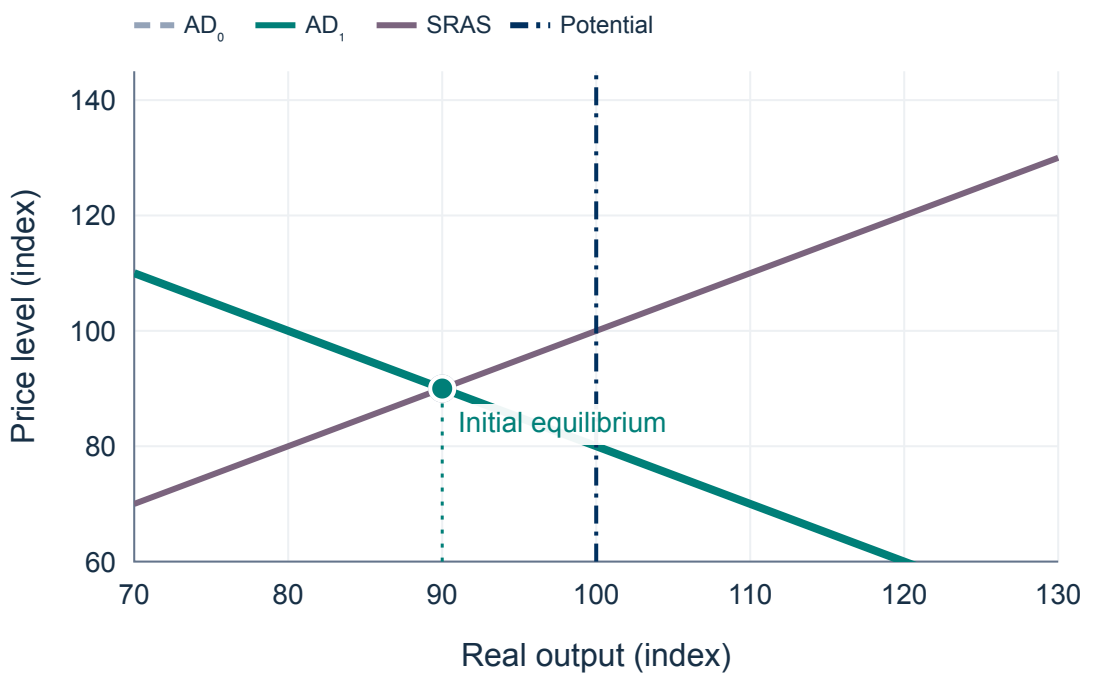
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G: Government purchases	Government buys current goods and services. The direct expenditure channel.
NX: Net exports	Higher domestic spending can increase imports. Imports reduce NX, other things equal.

Expansionary Fiscal Policy

Starting output = 90 · Potential output = 100

Expansionary · AD increases



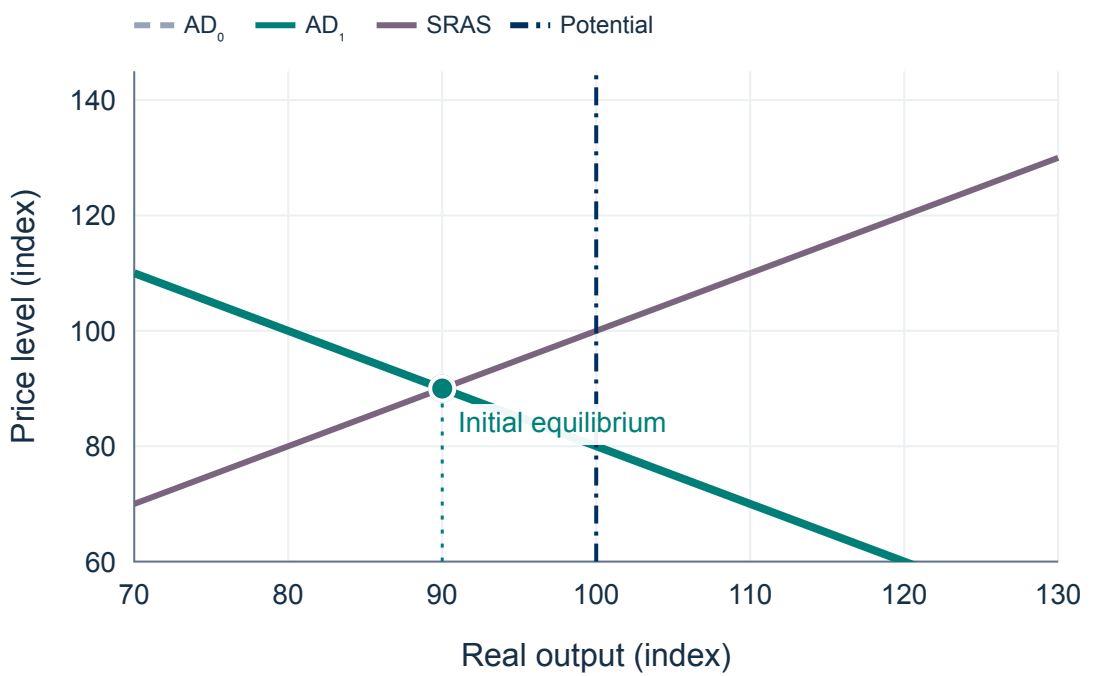
Recessionary gap

Output is below potential.

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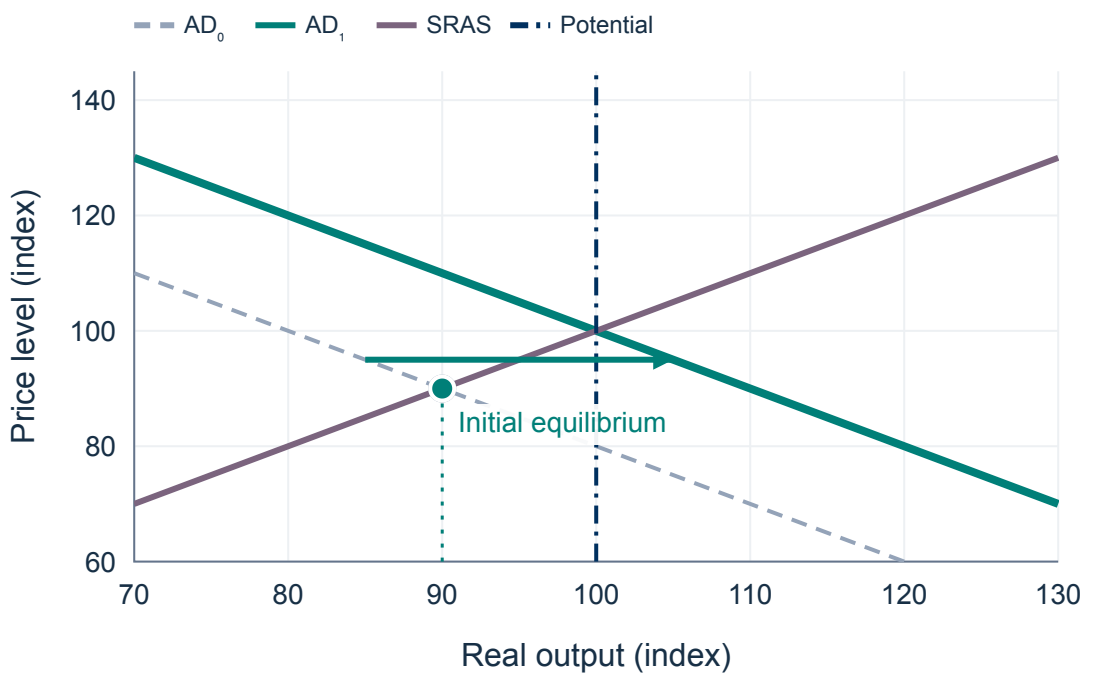
Policy change

Government purchases increase.

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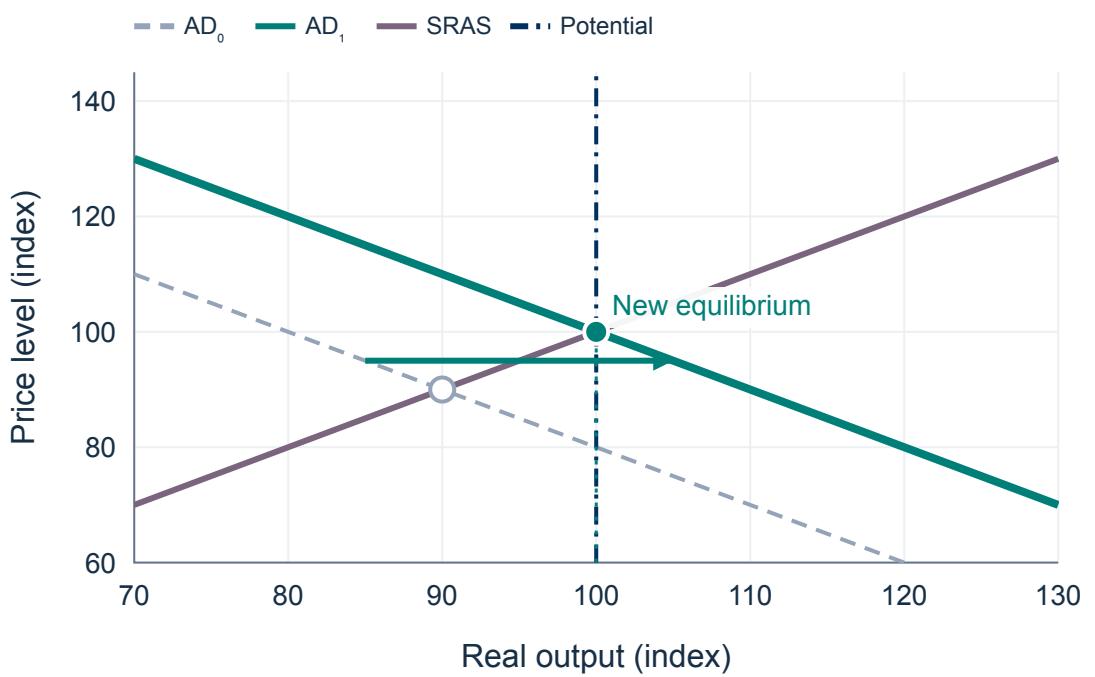
Demand response

AD shifts right; production rises.

Expansionary Fiscal Policy

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Recessionary gap

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Policy change

Government purchases increase.

Demand response

AD shifts right; production rises.

New equilibrium

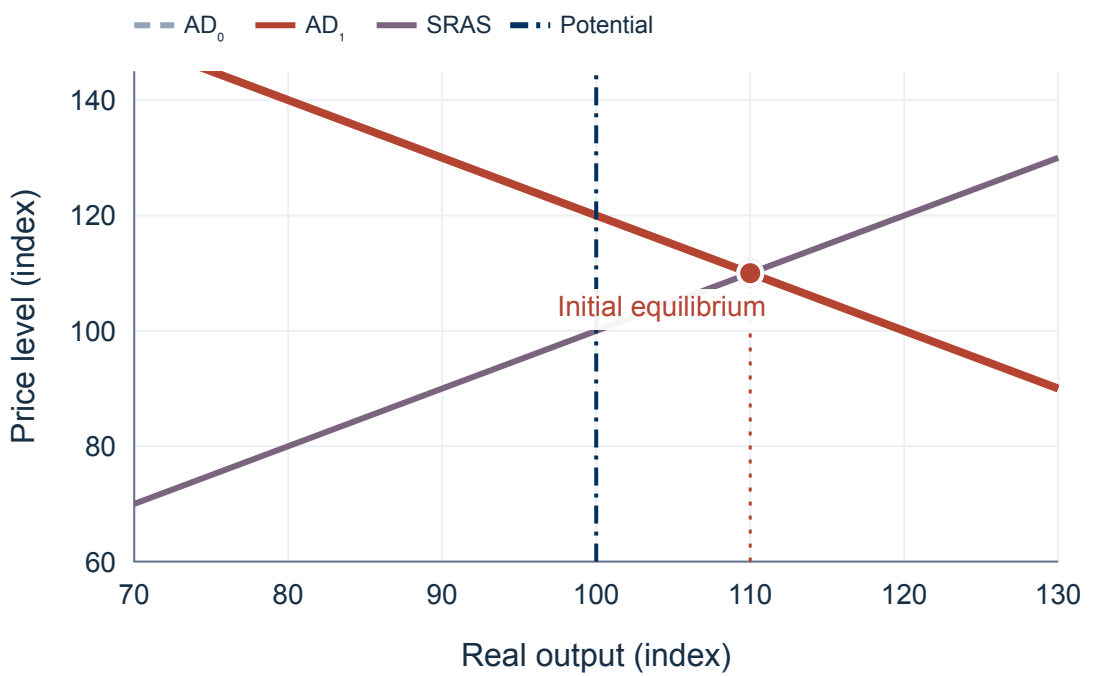
Output reaches 100; the price level rises.

Expansionary alternatives: G ↑, household T ↓, or TR ↑.

Contractionary Fiscal Policy

Starting output = 110 · Potential output = 100

Contractionary · AD decreases



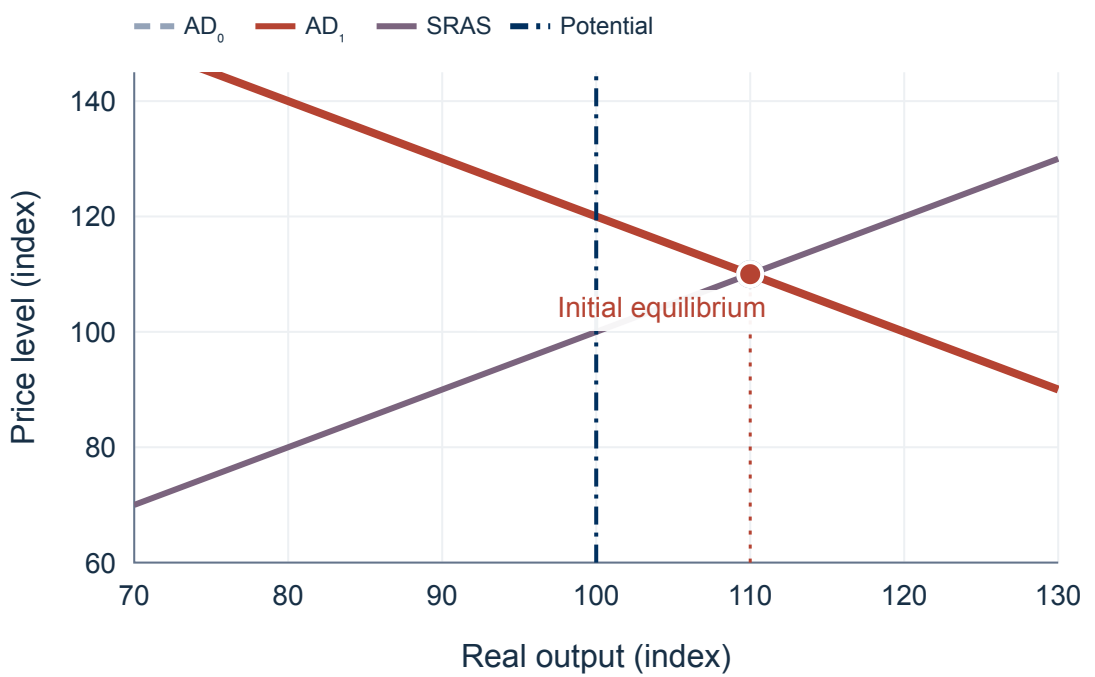
Inflationary gap

Output is above potential.

Contractionary Fiscal Policy

Starting output = 110 · Potential output = 100

Contractionary · AD decreases



Inflationary gap

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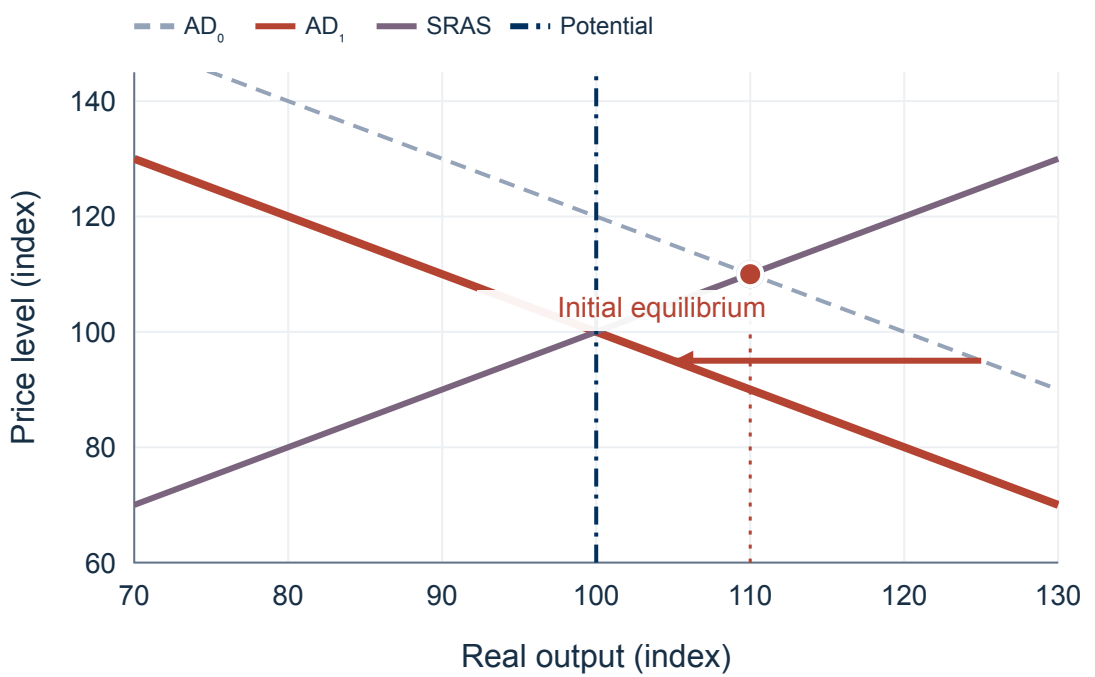
Policy change

Government purchases decrease.

Contractionary Fiscal Policy

Starting output = 110 · Potential output = 100

Contractionary · AD decreases



Inflationary gap

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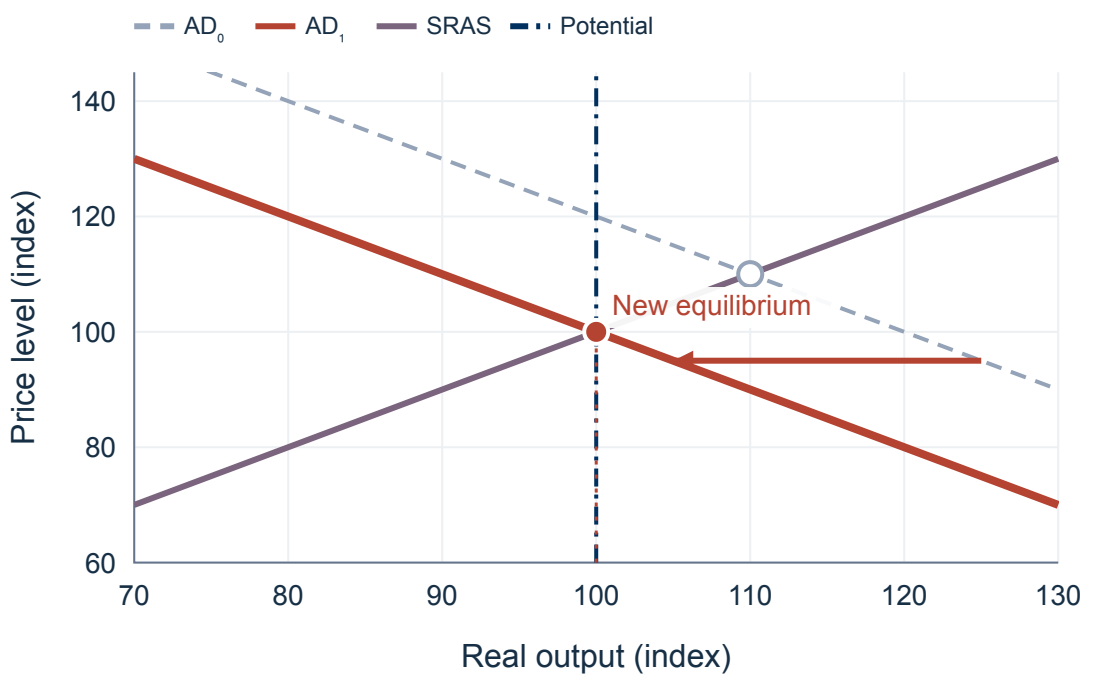
Demand response

AD shifts left; production falls.

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Inflationary gap

Output is above potential.

Policy change

Government purchases decrease.

Demand response

AD shifts left; production falls.

New equilibrium

Output returns to 100; the price level falls.

Contractionary alternatives: G ↓, household T ↑, or TR ↓.

Fiscal Policy Checkpoint

Identify the policy direction and the immediate spending channel.

A. Output is below potential; government repairs more roads.

B. Output is above potential; household income taxes rise.

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Expansionary. Purchases directly increase G.

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Contractionary. Disposable income and consumption fall.

Three Criticisms of Fiscal Expansion

Criticism	Mechanism	Qualification
Investment crowding out	Borrowing can raise interest rates and reduce I.	The size depends on economic and monetary conditions.
Consumption offset		
Ricardian equivalence		

An offset can weaken a policy without eliminating its effect.

References: CBO fiscal-policy framework; CRS fiscal-policy review

Three Criticisms of Fiscal Expansion

Criticism	Mechanism	Qualification
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Consumption offset	Current tax financing can reduce disposable income and C.	The reduction need not cancel the extra G.
Ricardian equivalence	A tax cut may be saved for expected future taxes.	A conditional tax-timing argument, holding spending fixed.

An offset can weaken a policy without eliminating its effect.

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Fiscal Policy Lags: The 2008–2009 Response

Recognition

Diagnose the shock.

15 Sep 2008

Lehman bankruptcy

26 Jan 2009

Recovery bill introduced

17 Feb 2009

ARRA signed

May 2009

$250 benefit payments

Authorization and delivery occur at different times.

Sources: Federal Reserve History; Congress.gov (H.R. 1); Social Security Administration

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The Spending Multiplier

> **The first payment becomes income**
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A government purchase pays a worker or supplier.

Some income is spent again

Recipients consume part and save part of the extra income.

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Further rounds become smaller

New consumption creates more income; saving limits the chain.

The output change combines the first spending change and the later rounds.

Government Purchases: Successive Spending Rounds

ΔG = +$100 billion · MPC = 0.8 · Fixed price level

Government purchases

+$100 billion

The initial purchase is income for workers and suppliers.

First spending

$100 billion

Fixed taxes and transfers; no imports or investment offsets.

Government Purchases: Successive Spending Rounds

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Next consumption round

0.8 × $100 = $80 billion

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Further consumption rounds

$64 billion + $51.2 billion + …

Total ΔY = $500 billion $k_G = 1/(1 - 0.8) = 5$

Fixed taxes and transfers; no imports or investment offsets.

A Household Tax Cut

Household taxes fall by $100 billion · MPC = 0.8

> **Disposable income rises**
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> $\Delta T = -\$100$ billion; $\Delta Y_D = +\$100$ billion.

A Household Tax Cut

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$\Delta T = -\$100$ billion; $\Delta Y_D = +\$100$ billion.

Households consume part

Initially, consumption rises $80 billion and saving rises $20 billion.

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Initially, consumption rises $80 billion and saving rises $20 billion.

Later rounds add to output

$\Delta Y = \$80 \text{ billion} \times 5 = +\$400$ billion.

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$\Delta Y = \$80 \text{ billion} \times 5 = +\$400$ billion.

The tax multiplier has a negative sign

$k_T = -4$, so $\Delta Y = (-4) \times (-\$100 \text{ billion}) = +\$400$ billion.

Purchases, Tax Cuts, and Transfers

MPC = 0.8 · Change one instrument at a time

Expansionary change	First channel	Initial expenditure
G increases by $100 billion	Government purchases	+$100 billion in G
Household T falls by $100 billion		
TR increases by $100 billion		

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TR increases by $100 billion	Disposable income +$100 billion	+$80 billion in C

Transfers affect GDP when recipients spend them; the payment itself is not G.

Fiscal Multipliers

MPC = 0.8 · Fixed price level · One policy change at a time

Expansionary change	First spending	Multiplier	Equilibrium ΔY
ΔG = +$100 billion	+$100 billion	$k_G = 5$	+$500 billion
ΔT = −$100 billion			
ΔTR = +$100 billion			

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ΔTR = +$100 billion			

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ΔTR = +$100 billion	+$80 billion	$k_{TR} = 4$	+$400 billion

$k_G = 1/(1 - MPC)$ $k_T = -MPC/(1 - MPC)$ $k_{TR} = MPC/(1 - MPC)$

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$k_G = 1/(1 - MPC)$ $k_T = -MPC/(1 - MPC)$ $k_{TR} = MPC/(1 - MPC)$

A $500 billion gap: ΔG = $500 billion ÷ 5 = $100 billion.

Recipients and Policy Design

An additional $100 of disposable income

Recipient	MPC	Initial consumption
Household A	0.8	$80
Household B	0.4	

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Recipients and Policy Design

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Consider spending, delivery speed, distribution, and the value of public purchases.

Income Taxes and the Multiplier

MPC = 0.8 · Income-tax rate t = 25% · Fixed price level

Additional output

→ →

$\Delta Y = \$100$

Income Taxes and the Multiplier

MPC = 0.8 · Income-tax rate t = 25% · Fixed price level

Additional output

$\Delta Y = \$100$

$\rightarrow$

Taxes and disposable income

$\Delta T = \$25 \quad \Delta Y_D = \$75$

$\rightarrow$

Income Taxes and the Multiplier

MPC = 0.8 · Income-tax rate t = 25% · Fixed price level

Additional output		Taxes and disposable income		Additional consumption
$\Delta Y = \$100$	$\rightarrow$	$\Delta T = \$25$ $\quad \Delta Y_D = \$75$	$\rightarrow$	$\Delta C = 0.8 \times \$75 = \$60$

Only \$60 reaches the next spending round, compared with \$80 under fixed taxes.

MPC remains 0.8; MPS remains 0.2.

Income Taxes and the Multiplier

MPC = 0.8 · Income-tax rate t = 25% · Fixed price level

Additional output		Taxes and disposable income		Additional consumption
$\Delta Y = \$100$	$\rightarrow$	$\Delta T = \$25$ $\Delta Y_D = \$75$	$\rightarrow$	$\Delta C = 0.8 \times \$75 = \$60$

Only \$60 reaches the next spending round, compared with \$80 under fixed taxes.

MPC remains 0.8; MPS remains 0.2.

$k_G = 1 / [1 - MPC(1 - t)]$

$k_G = 1 / [1 - 0.8(0.75)] =$ **2.5**, compared with **5** under fixed taxes.

Automatic Stabilizers

Income falls

Tax payments fall under existing income-tax rules.

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Unemployment rises

More eligible workers receive existing benefits.

Automatic Stabilizers

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Consumption is cushioned

Disposable income falls by less than it otherwise would.

No new law is needed for these responses.

Government Budget Balance

$$S_g = T - G - TR$$

T: tax revenue G: purchases TR: transfers

Positive balance	Zero balance	Negative balance
Surplus	Balanced budget	Deficit

Simplified budget: debt interest and other budget items omitted.

Government Budget Balance

$$S_g = T - G - TR$$

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Surplus	Balanced budget	Deficit

$T = \$500\text{bn} \cdot G = \$400\text{bn} \cdot TR = \$150\text{bn}$

$S_g = \$500\text{bn} - \$400\text{bn} - \$150\text{bn} = \mathbf{-\$50\text{bn}}$

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The balance alone does not reveal the effect on output.

Simplified budget: debt interest and other budget items omitted.

Balanced-Budget Expansion

ΔG = +$100bn · ΔT = +$100bn · ΔTR = $0 · MPC = 0.8

Spending channel	Change ($ billions)
Government purchases	+100
Initial consumption	
Initial net expenditure	

Fixed lump-sum taxes, fixed prices, unchanged transfers; no imports or interest/investment offsets.

Balanced-Budget Expansion

$\Delta G = +\$100bn \cdot \Delta T = +\$100bn \cdot \Delta TR = \$0 \cdot MPC = 0.8$

Spending channel	Change ($ billions)
Government purchases	+100
Initial consumption	−80
Initial net expenditure	+20

Fixed lump-sum taxes, fixed prices, unchanged transfers; no imports or interest/investment offsets.

Balanced-Budget Expansion

$\Delta G = +\$100bn \cdot \Delta T = +\$100bn \cdot \Delta TR = \$0 \cdot MPC = 0.8$

Spending channel	Change ($ billions)
Government purchases	+100
Initial consumption	−80
Initial net expenditure	+20

$\Delta Y = 5(100) - 4(100) =$ **+$100 billion**

Fixed lump-sum taxes, fixed prices, unchanged transfers; no imports or interest/investment offsets.

Balanced-Budget Expansion

$\Delta G = +\$100bn \cdot \Delta T = +\$100bn \cdot \Delta TR = \$0 \cdot MPC = 0.8$

Spending channel	Change ($ billions)
Government purchases	+100
Initial consumption	−80
Initial net expenditure	+20

$\Delta Y = 5(100) - 4(100) =$ **+$100 billion**

$\Delta S_g = 100 - 100 - 0 =$ **$0**

The balanced-budget multiplier is 1; the balance is unchanged.

Fixed lump-sum taxes, fixed prices, unchanged transfers; no imports or interest/investment offsets.

The Budget and the Business Cycle

Existing tax schedules and benefit rules · No new legislation

Budget item	Activity weakens	Activity strengthens
Tax revenue	Usually falls	Usually rises
Unemployment benefits		
Budget balance		

Concept reference: CBO, Effects of Automatic Stabilizers on the Federal Budget (2024)

The Budget and the Business Cycle

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Budget balance		

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The Budget and the Business Cycle

Existing tax schedules and benefit rules · No new legislation

Budget item	Activity weakens	Activity strengthens
Tax revenue	Usually falls	Usually rises
Unemployment benefits	More eligible claims	Fewer eligible claims
Budget balance	Moves toward deficit	Moves toward surplus

Moving toward surplus can mean a smaller deficit.

Concept reference: CBO, Effects of Automatic Stabilizers on the Federal Budget (2024)

Cyclically Adjusted Budget Balance

Estimated balance under current policies with real GDP at potential output.

Measure	Balance ($ billions)
Actual balance in recession	−150
Estimated balance at potential	
Estimated cyclical contribution	

Potential output and cyclical responses are estimated.

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Cyclically Adjusted Budget Balance

Estimated balance under current policies with real GDP at potential output.

Measure	Balance ($ billions)
Actual balance in recession	−150
Estimated balance at potential	−50
Estimated cyclical contribution	

Potential output and cyclical responses are estimated.

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Cyclically Adjusted Budget Balance

Estimated balance under current policies with real GDP at potential output.

Measure	Balance ($ billions)
Actual balance in recession	−150
Estimated balance at potential	−50
Estimated cyclical contribution	−100

Actual = adjusted + cyclical

−150 = −50 + (−100)

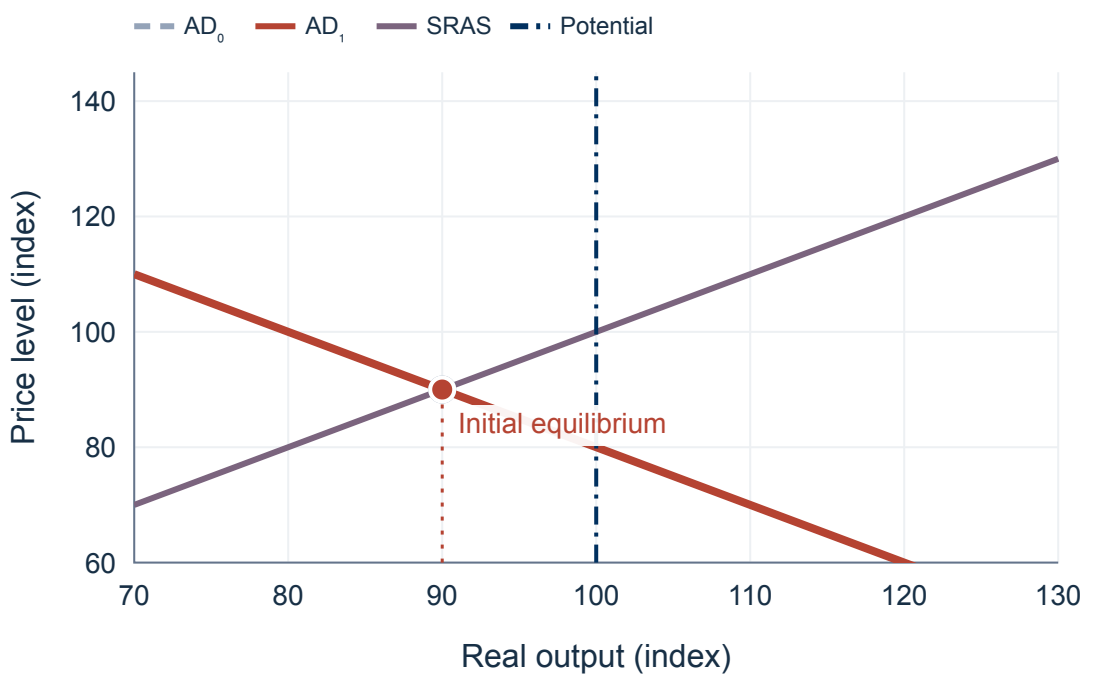
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Annual Budget Balance and Stabilization

Demand recession · Starting output = 90 · Potential output = 100

Contractionary · AD decreases



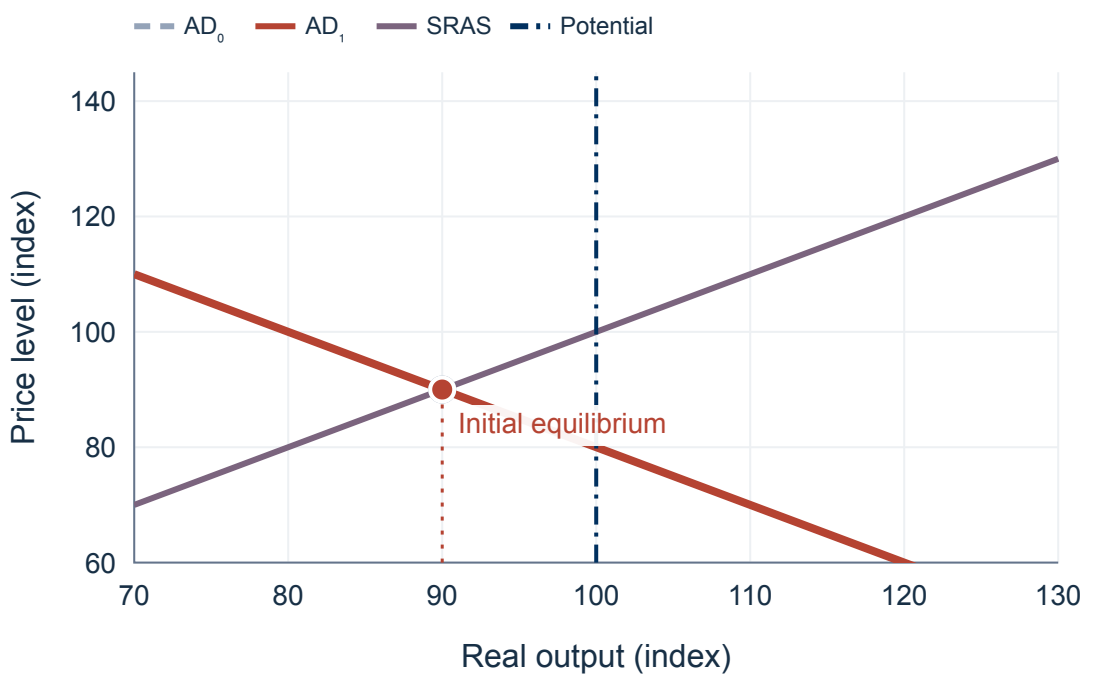
Initial recession

Output is already below potential.

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Contractionary · AD decreases



Initial recession

Output is already below potential.

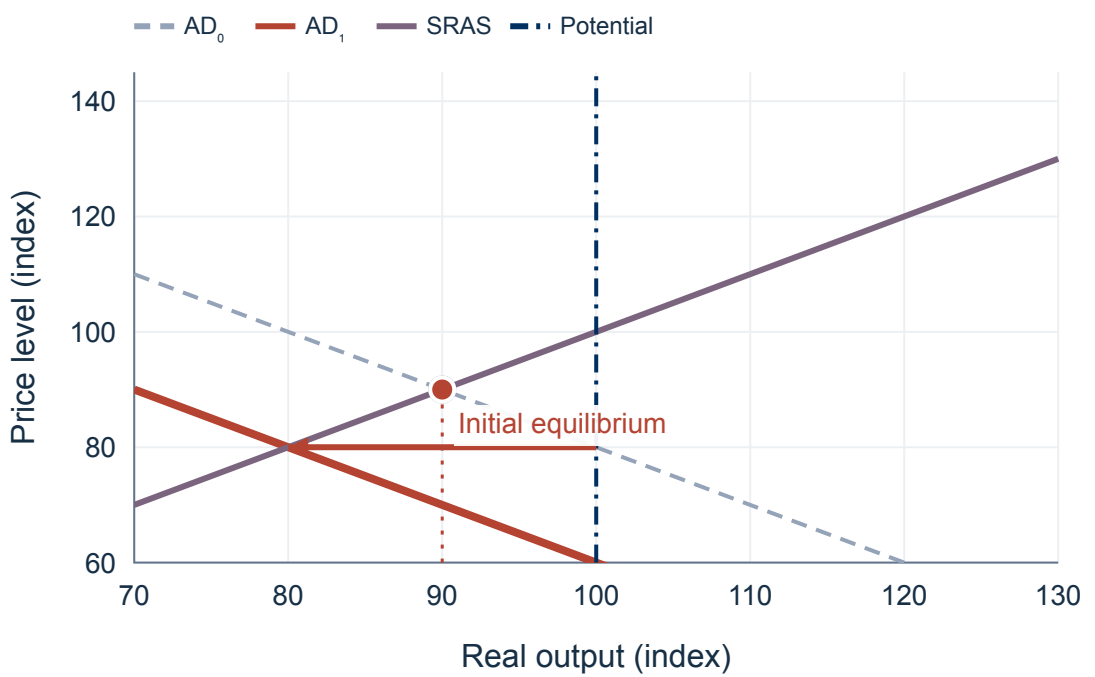
Annual-balance requirement

Falling receipts trigger tax increases or spending cuts.

Annual Budget Balance and Stabilization

Demand recession · Starting output = 90 · Potential output = 100

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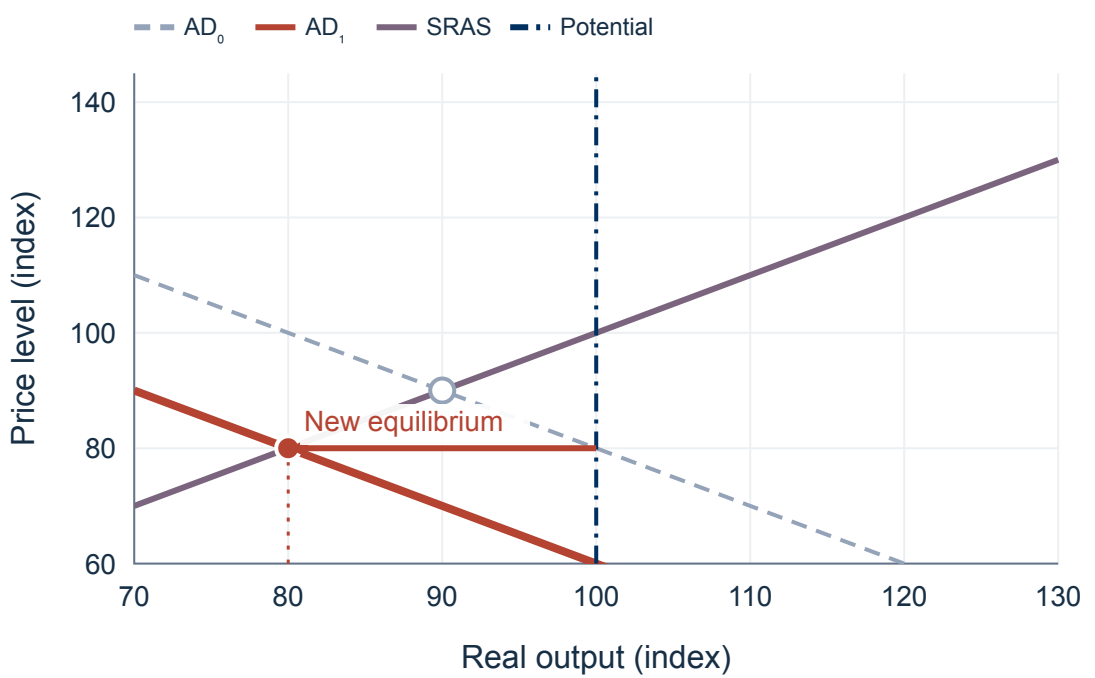
Demand response

The offset reduces AD further.

Annual Budget Balance and Stabilization

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Deeper output gap

Output falls from 90 to 80.

A strict annual-balance rule can work against automatic stabilizers.

Fiscal Stabilization and Sustainability

Downturns

Fiscal space allows stabilizers and temporary support to operate.

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Debt over time

Assess debt relative to GDP, interest costs, growth, and future balances.

A deficit is an annual flow; debt is a stock accumulated over time.

Fiscal Policy: Final Checkpoint

Explain the mechanism and the evidence you would need.

A. Income falls; the tax schedule stays unchanged.

B. A new law raises weekly unemployment benefits.

C. With MPC = 0.8, compare a $100 purchase and a $100 tax cut.

D. The deficit rises in a recession. Does this prove new stimulus?

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Check new laws, automatic stabilizers, and the cyclically adjusted balance.

Fiscal Policy Cheat Sheet: Demand

Expenditure: C + I + G + NX. Purchases enter G; household taxes and transfers influence C; business taxes can influence I.

Policy	Main mechanism	Curve	Short-run result
Increase purchases	Government buys more current goods and services.	AD right	Output and price level rise.
Cut household taxes	Disposable income rises; households spend part.	AD right	Output and price level rise.
Increase transfers	Recipients spend part of the extra income.	AD right	Output and price level rise.
Cut purchases or transfers; raise household taxes	Spending falls.	AD left	Output and price level fall.

First three: expansionary. Fourth: contractionary. Transfers themselves are not G.

Accounting: BEA · Demand mechanism: CBO

Fiscal Policy Cheat Sheet: Supply

Fiscal policy can also affect firms' costs and the economy's productive capacity.

Policy	Main mechanism	Possible curve shifts	Conditional result
Productive infrastructure, education or research	Purchases support demand now; successful investment improves capacity later.	AD right initially; possibly SRAS and LRAS right later.	Potential output can rise.
Lower production taxes or subsidize inputs	Lower marginal production costs can encourage supply.	Potentially SRAS right.	Output rises and price level falls from the supply effect alone.

Design and financing matter. Public spending must improve productivity to raise capacity.

Fiscal framework: CBO, How CBO Analyzes Fiscal Policies

Demand Shocks: Adjustment and Policy

Y* = potential output. Automatic adjustment below assumes no policy response and unchanged potential.

Initial shock	Immediate result	Automatic adjustment	Fiscal response to the gap	Policy consequence
AD left	Y < Y*; price level falls.	Wages and costs fall; SRAS moves right.	Expansionary policy moves AD right.	Supports output and offsets some downward price pressure.
AD right	Y > Y*; price level rises.	Wages and costs rise; SRAS moves left.	Contractionary policy moves AD left.	Returns output toward potential and eases price pressure.

Self-adjustment changes SRAS through wages and costs; fiscal stabilization usually changes AD.

Supply Shocks: Adjustment and Policy

Y* = potential output. These two cost shocks leave potential unchanged; adjustment assumes no policy response.

Initial shock	Immediate result	Automatic adjustment	Fiscal response to the gap	Policy consequence
SRAS left	Y < Y*; price level rises.	Lower wages or reversal of the cost shock can move SRAS right.	Expansionary policy moves AD right.	Supports output, but raises the price level further.
SRAS right	Y > Y*; price level falls.	Higher wages or reversal of the favorable shock can move SRAS left.	Contractionary policy moves AD left.	Returns output toward potential, but lowers the price level further.

After an adverse supply shock, demand policy faces an output–price tradeoff.

Potential Output and Policy Tradeoffs

Productive capacity changes

LRAS shifts; SRAS may also shift. Adjustment is toward the new potential output.

Choose the relevant benchmark

Fiscal choices depend on the remaining gap and the policy objective. Restoring the old GDP level may be inappropriate.

Keep the two mechanisms distinct

Wage and cost adjustment shifts SRAS; fiscal stabilization usually shifts AD.

Results depend on timing, financing, household spending, productivity, and central-bank responses.